

VietProfs: Building a Self-Maintaining Directory of the Vietnamese Academic Diaspora

Anonymous authors for review

August 2026

Abstract

Public directories of specialized populations are deceptively difficult to keep accurate. The records describe people and institutions on the open web, where appointments, titles, affiliations, and pages change without a shared update protocol. We present VietProfs, a deployed static web directory of Vietnamese and Vietnamese-diaspora faculty at universities outside Vietnam, as an experience report on this problem. Its workflow combines bounded search tasks, agent-assisted candidate generation and revalidation, structured proposals, deterministic validation, a tracked verification ledger, and human or independent review for ambiguous cases. The central design rule is operational rather than aspirational: AI proposes; evidence decides. As of 29 August 2026, the canonical snapshot contains 831 records at 367 universities in 19 countries, organized into 17 broad fields and five appointment tracks. Runtime aggregation exposes institutional and disciplinary patterns, while Git history records additions, promotions or track corrections, moves, removals, and deduplication. We distinguish expansion—finding people not yet represented—from revalidation—checking existing claims against the changing web. The system is not a complete census and does not provide controlled accuracy or cost estimates. Its contribution is a concrete engineering pattern: a public directory is a continuously revalidated collection of claims about an external world, not merely a database snapshot.

1 Introduction

Structured public datasets become stale for a simple reason: the world they describe changes while the dataset waits for an update. A faculty member is promoted, changes universities, becomes emeritus, leaves academia, or acquires a new official profile. A URL may disappear during a site redesign. A newly eligible person may never appear in the search results that originally seeded a directory. Each row is consequently a claim about an external entity, and maintenance is a reconciliation problem between those claims and a distributed, changing web.

VietProfs is a public directory of Vietnamese and Vietnamese-diaspora professors at universities worldwide. The site is intentionally small in data-engineering terms—a static Vite application whose canonical roster is JSON—but the population it represents is open-world, heterogeneous, and not available from a single authoritative source. Ordinary search is insufficient because faculty pages are inconsistently indexed, academic titles vary by country, and a plausible name or research mention is not proof of a current appointment. Manual curation alone is difficult to sustain as the directory grows.

This paper uses VietProfs as a case study for a broader question: how can AI-assisted automation continuously discover, verify, correct, analyze, and maintain a public directory while retaining evidence and human oversight? We make four bounded contributions. First, we document the data model and operational architecture of a live directory. Second, we describe a workflow that separates candidate discovery from evidence-backed verification. Third, we characterize expansion and revalidation using implementation artifacts and public Git history. Fourth, we show how a maintained roster supports reproducible descriptive observations while also creating a second maintenance problem for derived knowledge.

2 VietProfs and the Data Model

The public source of truth is `public/data.json`; the browser loads it and performs search, filtering, sorting, and aggregation. A record contains a name, current profile URL, university, city, state or province when applicable, country, department, research areas, rank and appointment track, optional education and honors, optional Scholar and personal-site URLs, portrait provenance, and `lastUpdatedAt`. The separate `maintenance/verification.json` ledger records the time of a completed full review and is not part of the public site payload.

The inclusion rule is a classification policy. A person must have a current primary university appointment outside Vietnam, except for qualifying emeritus appointments, and must fit one of five tracks: Tenure-line, Teaching, Research, Clinical, or Emeritus. Adjunct, visiting, postdoctoral, affiliate/courtesy, temporary, part-time, non-university, and industry-only positions are excluded. Teaching includes stable full-time non-tenure-track appointments such as Professor of Practice; Research and Clinical require evidence that the appointment is faculty-level and stable; Emeritus requires a formally conferred title following a tenure-line career. The published rank is normalized for the public vocabulary where appropriate, while institution-specific titles are preserved when they clarify a non-tenure appointment. [CODE: `ROSTER_MAINTENANCE.md`; `src/data.ts`]

Vietnamese identity is not inferred from a surname. Vietnamese names and common surnames are used as search signals, but eligibility depends on the appointment and the project’s inclusion policy. This distinction matters for both ethics and correctness: `Nguyen` is neither sufficient evidence of identity nor a reliable duplicate key. Names are deduplicated by person, with a university suffix used only when genuinely identical names require disambiguation.

Departments are mapped to 17 broad fields by ordered rules and institution-specific overrides. Health Sciences additionally has derived subfields such as Clinical Medicine, Public Health, Nursing, Pharmacy, Dentistry, and Biomedical Research. The mapper is deliberately explicit because strings such as “Information Studies” or “Mathematics and Statistics” can mean different things in different institutional contexts. The tests assert both the taxonomy and representative overrides. [CODE: `src/data.ts`; `test/data.test.ts`]

3 Why Maintaining the Directory Is Hard

The task has five coupled sources of uncertainty. First, it is open-world: the set of eligible people is not enumerated anywhere, so absence from a search result is weak negative evidence. Second, official sources are heterogeneous. A department directory, a university news article, a CV, a personal homepage, and a conference biography have different roles and freshness. Third, academic titles are not globally interchangeable; a research assistant professor can be a career research rank at one institution and a temporary postdoctoral-like role at another. Fourth, identity resolution is hard when names are common, transliterated, reordered, accented, abbreviated, or shared by multiple people. Finally, web pages are mutable: a 404 may indicate a moved page or site redesign rather than a departure.

These conditions make a row better understood as a claim such as “person p currently holds appointment a at institution u , supported by source s ,” with a future revalidation obligation. The database is a materialized view of claims; the sources and review history are what make the view maintainable.

4 System Architecture

Figure 1 summarizes the implemented flow. Public web sources feed either bounded discovery or a targeted revalidation task. An agent returns a structured proposal and report. The controller applies deterministic schema and scope checks before changing the canonical roster. The static site is built from the resulting snapshot, while the same snapshot is later selected for revalidation using the ledger.

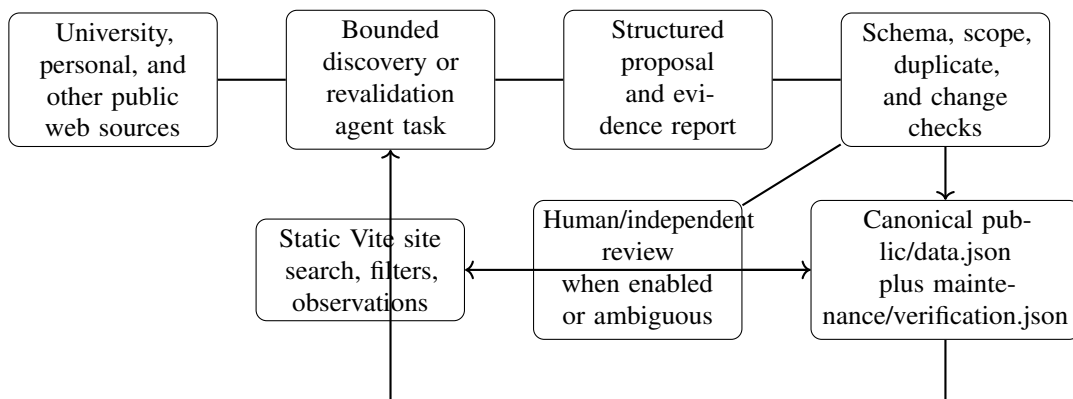


Figure 1: VietProfs architecture. Solid arrows denote implemented data flow; the lower loop represents periodic revalidation.

4.1 Controller and proposal boundary

`scripts/maintain-roster.ts` is the unattended maintenance controller. It normally selects entries missing from the ledger or whose last full review is at least 365 days old, prioritizing older and incomplete records, and processes at most 40 per new run. It supports dry runs, full or capped sweeps, name/field targeting, resumable checkpoints, safe stop/status, retry handling, and Claude as the default research agent. Codex can be selected as the agent or enabled as an independent review pass.

Agents cannot edit repository files directly. They return schemas for target matching, research, and optional independent review. The controller validates required fields, HTTP(S) roles, accepted tracks, research areas, honors and degree shapes, portrait paths, and other invariants. It also checks that a proposal changes only the targeted record, does not reorder the roster, and does not silently modify unrelated entries. A changed public record receives a new `lastUpdatedAt`; a complete review that finds no substantive change advances only the ledger timestamp. [CODE: `scripts/maintain-roster.ts`; `test/maintenance-controller.test.ts`]

The deployment workflow runs tests and a production build on every push to `main`, then deploys the static artifact to GitHub Pages. The maintenance controller can commit and push after checks. This is a strong degree of operational automation, but it is not a formal correctness guarantee and does not eliminate human judgment for ambiguous evidence.

5 Discovery as Partitioned Open-World Search

An instruction such as “find all Vietnamese professors at universities worldwide” has unbounded scope and unmeasurable recall. VietProfs instead generates repeatable queries for a bounded university, field, and domain, combining site-restricted searches, department/faculty terms, institutional news and research-center queries, and Vietnamese-name signals. The agent is expected to audit existing entries, gather evidence per candidate, and treat coauthors, student pages, grants, and research mentions as leads rather than proof. [CODE: `scripts/faculty-discovery-queries.ts`; DOC: `ROSTER_MAINTENANCE.md`]

This decomposition is a practical form of bounded open-world search. It improves auditability: a research pass can say which university/field/country slice it attempted, what was added, and in some cases that no eligible addition was found. It does not solve recall. Search-engine ranking, English-language visibility, uneven university websites, and differing effort across slices remain sampling mechanisms. The repository history shows a switch to country-by-country searches, batches for international universities, and field audits. These are operational strategies, not evidence of exhaustive coverage.

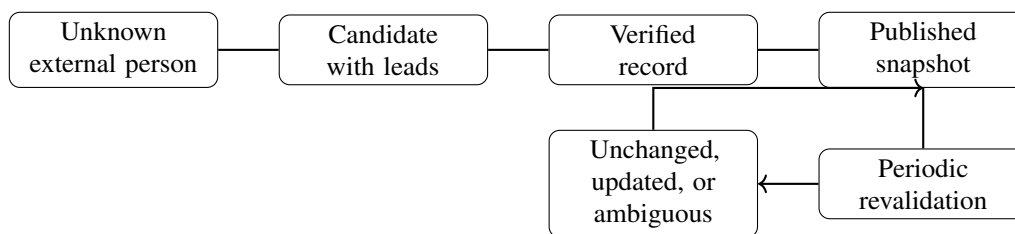


Figure 2: Record lifecycle. A published record is not terminal; it returns to the revalidation loop.

6 Evidence and Verification

Discovery is not verification. A candidate enters the workflow because a query or lead suggests a person; publication requires evidence that the person is current, at the claimed institution and department, in an eligible track, with a working official profile. Official faculty pages are the preferred source. Personal pages and CVs can supply details, but the current appointment is still confirmed on an official source where possible. Search snippets, conference biographies, and coauthorship are leads or corroboration rather than default authority.

The implementation therefore follows the rule “AI proposes; evidence decides.” The agent can read public pages, compare an old record with current evidence, and produce structured data, but the controller does not treat free-form model text as canonical state. The proposal schema requires an action, status, report, sources, and proposed JSON; an optional independent Codex review can approve, reject, or mark uncertain. Rejected proposals receive bounded revisions in the configured workflow; incomplete or uncertain work is deferred rather than marked successfully verified.

The evidence model is practical rather than a formal confidence score. URLs are retained for the profile, honors, portrait, and optional personal/Scholar pages, and Git preserves the resulting patch. There is no per-field provenance graph or source snapshot store. This is an important limitation: a URL is evidence access, not a cryptographic proof that the page had a particular content at review time.

7 Continuous Revalidation and Self-Maintenance

VietProfs separates two loops, shown in Figure 2. Expansion starts from an unsearched slice and tries to add a new verified record. Revalidation starts from an existing record and checks whether its claim still matches the web. The latter checks profile liveness and role, URL-role correctness, Scholar identity, current rank/track, education, honors, and possible moves; it also looks for new candidates incidentally. The ledger allows the controller to choose the least recently reviewed records rather than repeatedly touching only newly added people.

Negative evidence requires caution. A disappeared faculty page can mean redesign, a moved URL, temporary failure, a university move, retirement, or departure. The maintenance policy explicitly avoids treating 404 as automatic deletion and calls for corroboration. Removal is also a stronger action than addition because it destroys a public claim and can erase a legitimate person from the directory. The system supports removal proposals, but the guide’s corroboration rule is an operational policy rather than a formally verified deletion protocol.

The Git history makes this lifecycle visible. A 20-batch full-roster refresh appears on 26 August 2026. Subsequent history records automated maintenance batches, country sweeps, a Teaching-to- Clinical correction, a duplicate merge, a university-name canonicalization, a corrected affiliation, and a policy-based removal. These examples show that maintenance is not a periodic cosmetic edit; it changes eligibility, identity, affiliation, and derived data assumptions.

Table 1: Current VietProfs snapshot and evidence-field coverage.

Records	831	Universities	367
Countries	19	U.S./international	554/277
Broad fields	17	Accepted tracks	5
Profile URLs	831	Scholar URLs	325
PhD institutions	656	PhD years	488
Honors-bearing records	178	PhD-year range	1962–2026

8 What the Directory Reveals

The site has a third function beyond building and maintaining knowledge: learning from the current roster. Primary knowledge is the individual record. Aggregation produces derived knowledge such as institutional concentration, field balance, geography, track distribution, awards, and doctoral institution patterns.

8.1 Runtime observations

The “Show me something interesting” view is computed at runtime from the current JSON snapshot. It reports U.S. and international counts, state/country groups, institution groups, same-institution same-field clusters, field comparisons, award summaries, PhD cohorts, and leaderboards. It stores no hand-written facts. Tests require multi-record, roster-only observations and reject unsupported surname, prestige, demographic, causal, or population claims. The result is therefore a descriptive view, not an estimate of the entire diaspora. [CODE: src/data.ts; src/main.ts; test/data.test.ts]

8.2 Snapshot observations

Table 1 summarizes the current snapshot. Within it, Monash University has 13 records, UC San Diego 12, and NUS 11. California and Texas contain 169 of 554 U.S. records (31%). The international subset contains 277 records across 18 non-U.S. country values. The four largest broad fields in the analysis are Computer and Information Sciences (132), Engineering (124), Business and Economics (112), and Health Sciences (99). Nanyang Technological University has six engineering records; the University of Minnesota has six health-science records; and the University of Technology Sydney has six engineering records. These are useful because they reveal institutional and disciplinary structure without claiming prestige, population size, or causation.

8.3 Derived knowledge also becomes stale

If a person moves, an institutional count changes. If a track is corrected, the track distribution changes. If a record is removed, a field comparison or same-institution cluster may no longer hold. The current implementation recomputes the visible observations from the live roster at render time, so the runtime view does not preserve stale fact text. It does not maintain a separate dependency graph for persisted derived facts because those facts are not persisted. A future system that stores notable findings would need dependencies, recomputation, and invalidation policies just as primary records do.

9 Operational Experience and Evaluation

This work is an operational characterization, not a controlled experiment. The repository provides strong evidence about implementation and history but not precision, recall, cost, or detection latency. The current snapshot has 701 Tenure-line, 68 Teaching, 37 Emeritus, 21 Clinical, and 4 Research records. It contains 103 personal/lab URLs, 325

Table 2: Representative public maintenance events in Git history.

Case	Repository evidence	Result
Duplicate	2aa15c0; University of Dayton	Duplicate entries for Tam V. Nguyen merged.
Policy removal	9a4a171; inclusion-standard decision	Nhung Nguyen (UCSF) removed.
Track correction	ba71beb; BU Dental	Teaching changed to Clinical.
Affiliation correction	c1f4380	Nguyen-Truc-Dao Nguyen affiliation corrected.
Name/identity cleanup	c0c0603, 4212955	True duplicate, suffix, and university-name canonicalization fixed.

Scholar URLs, and 178 records with at least one honor. These figures describe fields present in the snapshot, not their correctness or completeness.

The event history nevertheless supports meaningful engineering observations. Batch commits make large work reviewable; checkpoints make long agent sessions resumable; tests encode invariants such as unique names, unique profiles, valid tracks, and targeted proposal scope; and the ledger separates “reviewed” from “changed.” The absence of candidate-level telemetry is equally important. There is currently no denominator for candidates considered, no rejection taxonomy, no measured human review rate, and no time-to-detection metric.

10 What Does Not Work

Unbounded LLM enumeration is a poor source of a canonical roster. It omits people, hallucinates affiliations, and cannot provide a defensible stopping condition. Surname inference is unsafe and ethically inadequate. Search snippets are stale and incomplete. Direct model writes would make it difficult to constrain scope, distinguish a proposed change from an approved one, or recover from a mistaken bulk edit. VietProfs instead requires structured proposals and targeted changes.

Other failures are subtler. A broad title such as “Research Assistant Professor” cannot establish eligibility by itself. A page disappearance is not a departure. A personal homepage is not the same source role as an official university profile. Department keywords misclassify ambiguous institutional contexts unless overrides are maintained. Finally, a current count cannot be described as growth or a migration path without versioned historical data and an explicit comparison. These lessons are implemented in policy, tests, and recent corrective commits, but they remain areas for ongoing measurement.

11 Limitations and Ethics

VietProfs is not a census. Coverage is affected by search effort, English-language discoverability, university web quality, public-profile availability, initial field emphasis, international title conventions, and the differing maturity of collection passes. The 554/277 U.S./international split must not be interpreted as a population ratio. Nor should current concentrations be interpreted as prestige, demographic size, causation, migration history, or research quality.

The directory handles sensitive identity-adjacent information. It should not infer ethnicity or national origin from names, and it should publish only public professional information necessary for the directory’s purpose. Portraits, honors, and education need source and scope rules. Public submission and correction mechanisms help people contest errors, but a future release should expose clear removal/contact policy, retention expectations, and an audit trail for disputed changes.

12 General Lessons

The case study suggests the following design principles for agent-maintained public datasets:

1. Use language models primarily for candidate generation, comparison, and research assistance; do not make them the source of truth.
2. Separate discovery, evidence collection, verification, proposal, and canonical application.
3. Attach source URLs and review timestamps to claims, while recognizing that URLs alone are not complete provenance.
4. Partition open-world search into bounded, repeatable tasks.
5. Treat removal and negative evidence as higher-risk than addition.
6. Keep candidate state separate from canonical state and constrain proposed edits to a target.
7. Reserve human attention for ambiguity, identity resolution, policy exceptions, and disputes.
8. Design telemetry before deployment: outcomes, reasons, cost, latency, and reviewer actions.
9. Recompute or invalidate derived knowledge when primary claims change.

Together these principles implement the operational slogan “AI proposes; evidence decides” without pretending that the web or the model is fully reliable.

13 Related Work

VietProfs is related to entity resolution, web extraction, scholarly knowledge graphs, and provenance-aware maintenance, but it is not a replacement for a general scholarly index. Surveys of entity resolution emphasize end-to-end blocking, matching, and data-quality pipelines over heterogeneous descriptions [2, 3]. Recent work studies LLMs for entity matching, including prompt sensitivity, structured explanations, and robustness to unseen entities [5]; another line explicitly treats model queries as a costed uncertainty-reduction resource [4]. VietProfs uses the practical implication—models can help compare and organize evidence—but does not claim their reported benchmark performance for this domain.

Knowledge-graph maintenance work formalizes incremental updates and provenance for asserted and derived facts [1]. VietProfs is simpler: JSON records, URLs, Git history, and runtime aggregation rather than an RDF graph or why-provenance engine. The conceptual connection is nevertheless direct: when a source claim changes, dependent answers may change.

OpenAlex provides a useful contrast as a large, open scholarly knowledge graph of works, authors, venues, institutions, and concepts [6]. VietProfs has a narrower population, more detailed appointment eligibility, and a public-directory interaction model. DBLP and ORCID are also important reference ecosystems for scholarly identity, but VietProfs’s engineering problem is the current faculty appointment and its evidence, not publication identity alone. The contribution here is therefore an experience report about a specialized, continuously revalidated directory and the operational boundary between agent assistance and canonical data.

14 Conclusion

Small specialized public datasets used to be expensive to maintain because every correction required continuous human labor against a changing web. VietProfs shows how that labor can be reduced with bounded agent tasks, structured proposals, deterministic invariants, provenance links, resumable revalidation, and human oversight. It also shows why automation is not the same as autonomy: the directory remains incomplete, sources conflict, titles are ambiguous, and derived knowledge can become stale.

The broader view is that a public directory is a continuously revalidated collection of claims about a changing external world. The useful system boundary is therefore not “LLM writes JSON,” but “agents propose evidence-backed changes to a constrained claim store, which is periodically reconciled with its sources.” As the cost of this

reconciliation falls, more specialized public knowledge resources may become practical to sustain—provided their authors measure coverage, retain provenance, support correction, and state clearly what the data does and does not establish.

References

- [1] Khalid Belhajjame and Mohamed-Yassine Mejri. Online maintenance of evolving knowledge graphs with rdfs-based saturation and why-provenance support. *Journal of Web Semantics*, 78:100796, 2023.
- [2] Vassilis Christophides, Vasilis Efthymiou, Themis Palpanas, George Papadakis, and Kostas Stefanidis. End-to-end entity resolution for big data: A survey. *arXiv preprint arXiv:1905.06397*, 2019.
- [3] Ahmed K. Elmagarmid, Panagiotis G. Ipeirotis, and Vassilios S. Verykios. Duplicate record detection: A survey. *IEEE Transactions on Knowledge and Data Engineering*, 19(1):1–16, 2007.
- [4] Huahang Li, Longyu Feng, Shuangyin Li, Fei Hao, Chen Jason Zhang, and Yuanfeng Song. On leveraging large language models for enhancing entity resolution: A cost-efficient approach. *arXiv preprint arXiv:2401.03426*, 2024.
- [5] Ralph Peeters, Aaron Steiner, and Christian Bizer. Entity matching using large language models. *arXiv preprint arXiv:2310.11244*, 2023.
- [6] Jason Priem, Heather Piwowar, and Richard Orr. Openalex: A fully-open index of scholarly works, authors, venues, and institutions. *arXiv preprint arXiv:2205.01833*, 2022.